Tut 6

Ex. 9.7

Q13 $\displaystyle\sum_{n=1}^{\infty} \frac{4^n x^{2n}}{n}$

Let $a_n = \dfrac{4^n x^{2n}}{n}$ for $n \geq 1$

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{4^{n+1}}{4^n} \cdot \frac{n}{n+1} \cdot \frac{|x|^{2n+2}}{|x|^{2n}}$$

$$= 4 \cdot \frac{n}{n+1} |x|^2$$

$$= 4 \cdot \frac{n}{n+1} x^2$$

$\lim \frac{n}{n+1} = \lim \frac{1}{1+\frac{1}{n}}$

$\therefore \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 4x^2$

$4x^2 < 1$ iff $x^2 < \frac{1}{4}$

iff $-\frac{1}{2} < x < \frac{1}{2}$

iff $|x| < \frac{1}{2}$ (optional)

For $-\frac{1}{2} < x < \frac{1}{2}$, $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$

$\Rightarrow \sum a_n$ conv. abs. by Ratio Test

For $x > \frac{1}{2}$ or $x < -\frac{1}{2}$, $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| > 1$

$\Rightarrow \sum a_n$ div. by Ratio Test.

Sort $x = -\frac{1}{2}$, $x = \frac{1}{2}$ Study the series

Q15 $\sum \frac{x^n}{\sqrt{n^2+3}}$

Let $a_n = \frac{x^n}{\sqrt{n^2+3}}$ for $n \geq 1$

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{\sqrt{n^2+3}}{\sqrt{(n+1)^2+3}} \cdot \frac{|x|^{n+1}}{|x|^n}$$

$$= |x| \cdot \frac{\sqrt{n^2+3}}{\sqrt{n^2+2n+4}}$$

$$\therefore \lim \left|\frac{a_{n+1}}{a_n}\right| = |x|$$

$$\lim \frac{\sqrt{n^2+3}}{\sqrt{n^2+2n+4}}$$

$$= \lim \frac{\sqrt{1+\frac{3}{n^2}}}{\sqrt{1+\frac{2}{n}+\frac{4}{n^2}}}$$

For $|x| < 1$, $\lim \left|\frac{a_{n+1}}{a_n}\right| < 1$

$\Rightarrow \sum a_n$ conv. abs. by Ratio Test

For $|x| > 1$, $\lim \left|\frac{a_{n+1}}{a_n}\right| > 1$

$\Rightarrow \sum a_n$ div. by Ratio Test

When $x = -1$, (Show $\sum \frac{(-1)}{\sqrt{n^2+3}}$ ~~is~~ conv. conditionally using AST)

When $x = 1$, (Show $\sum \frac{1}{\sqrt{n^2+3}}$ div. using LCT)